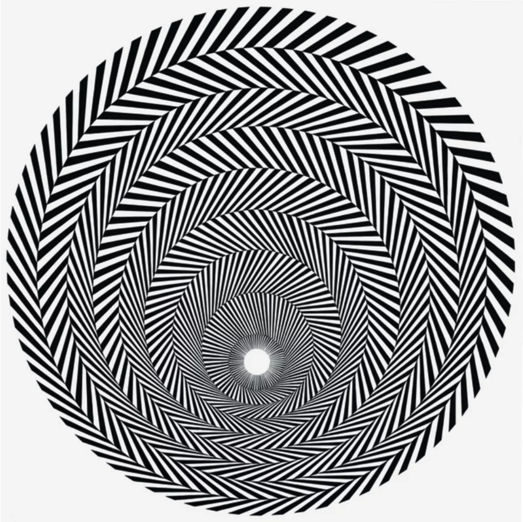


Bridget Riley (1964) Blaze 4

The Visual System

Primary visual cortex (V1)

Computational Neuroscience

University of Bristol

mrule

Learning outcomes:

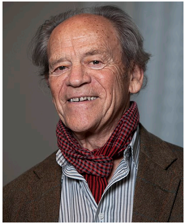
- Primary visual cortex (V1) tuning:
  - Simple cells
  - Complex cells
- The cortical microcircuit

Primary Visual Cortex (V1)

Hubel and Wiesel

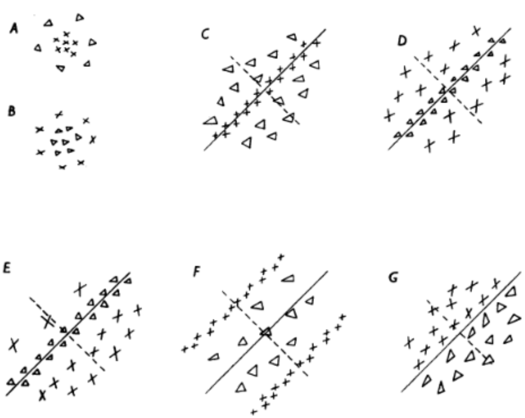
In 1959, David Hubel and Torston Wiesel place electrodes in primary visual cortex (V1) of a cat. They presented moving bars on a dark background, and grouped neurons they recorded from into **simple** and **complex** cells, based on their tuning properties.

David Hubel



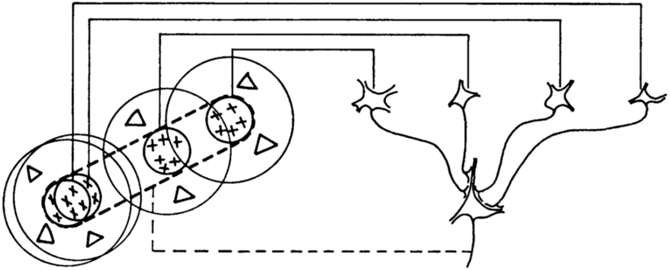
Torston Wiesel

Simple Cells

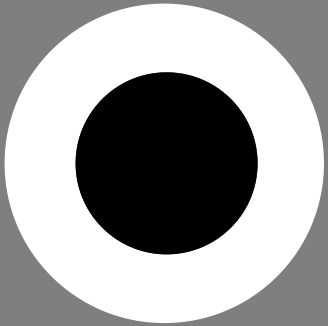


Text-fig. 2. Common arrangements of lateral geniculate and cortical receptive fields. *A*. 'On'-centre geniculate receptive field. *B*. 'Off'-centre geniculate receptive field. *C*–*G*. Various arrangements of simple cortical receptive fields. ×, areas giving excitatory responses ('on' responses); △, areas giving inhibitory responses ('off' responses). Receptive-field axes are shown by continuous lines through field centres; in the figure these are all oblique, but each arrangement occurs in all orientations.

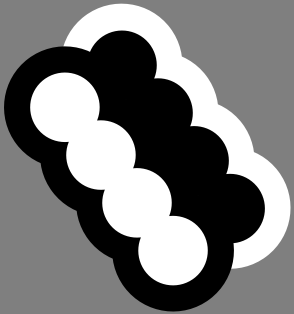
See video: Hubel and Wiesel (1962)

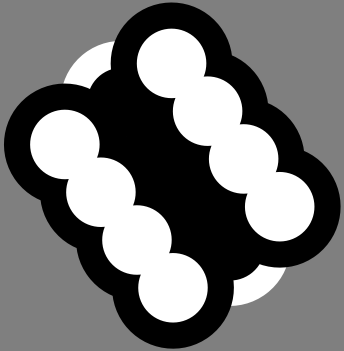


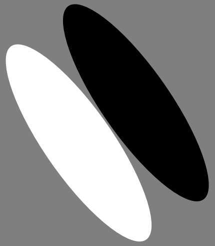
Text-fig. 19. Possible scheme for explaining the organization of simple receptive fields. A large number of lateral geniculate cells, of which four are illustrated in the upper right in the figure, have receptive fields with 'on' centres arranged along a straight line on the retina. All of these project upon a single cortical cell, and the synapses are supposed to be excitatory. The receptive field of the cortical cell will then have an elongated 'on' centre indicated by the interrupted lines in the receptive-field diagram to the left of the figure.

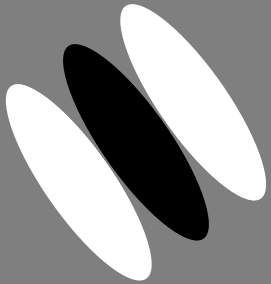


! On/off centre-surround contract cells are found in the retina (RGCs) and LGN





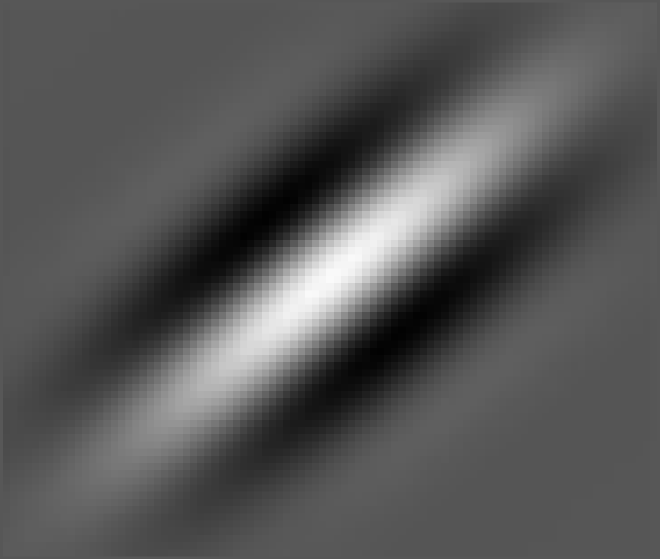


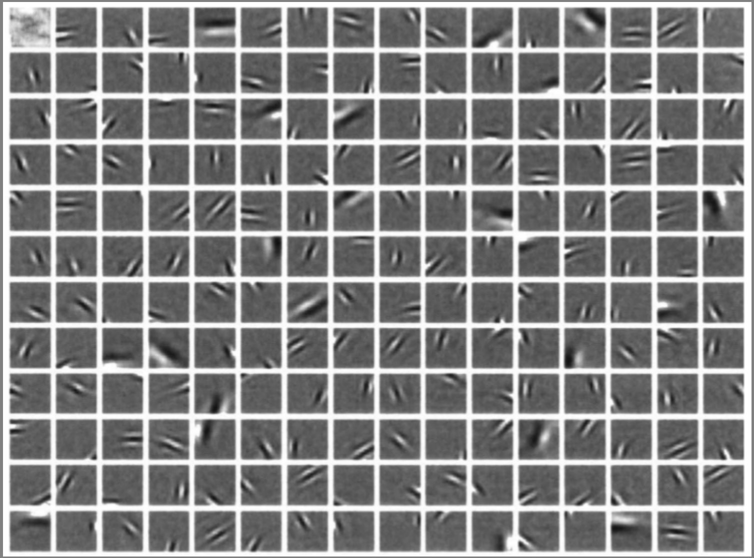


! **Simple cells are found in V1 layers 4 and 6 that receive direct input from LGN**

A **Gabor filter** is the product of a (2D) Gaussian with a sinusoidal plane wave, and provides a good approximation of V1 neuron's receptive fields (the Gaussian part), and orientation tuning properties (the plane wave part).

Gabor filter = Gaussian × sinusoid plane wave





Olshausen, Field (1996) Emergence of simple-cell receptive field properties by learning a sparse code for natural images

Hubel & Wiesel Simple Cells

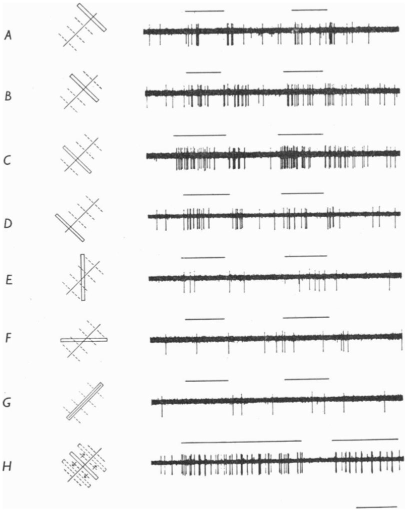
V1 simple cells $\sim$ ***Gabor filters***

$$\text{response} \sim \int_{\boldsymbol{x}} e^{-\frac{\|\boldsymbol{x}-\boldsymbol{x}_0\|^2}{\sigma^2}} \cdot \cos(\omega\, \hat{\boldsymbol{v}}^\top \boldsymbol{x} + \phi) \cdot u(\boldsymbol{x}), \qquad \text{where } u(\boldsymbol{x}) \sim (\text{light} - \text{dark}) \text{ image data}$$

$\exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{x}_0\|^2}{\sigma^2}\right)$	Localised, Gaussian ***receptive field*** near $\boldsymbol{x}_0$, size $\sim \sigma$
$\cos(\omega\, \hat{\boldsymbol{v}}^\top \boldsymbol{x} + \phi)$	Sinusoidal ***tuning*** $\sim$ edge $\parallel$ direction $\hat{\boldsymbol{v}}$, phase ϕ

- Found in V1, specifically layers 4 and 6 that receive input from thalamus

Complex Cells



Text-fig. 4. Responses of a cell with a complex field to stimulation of the left (contralateral) eye with a slit $\frac{1}{8} \times 2\frac{1}{2}°$. Receptive field was in the area centralis and was about $2 \times 3°$ in size. *A–D*, $\frac{1}{8}°$ wide slit oriented parallel to receptive field axis. *E–G*, slit oriented at 45 and 90° to receptive-field axis. *H*, slit oriented as in *A–D*, is on throughout the record and is moved rapidly from side to side where indicated by upper beam. Responses from left eye slightly more marked than those from right (Group 3, see Part II). Time 1 sec.

Hubel & Wiesel Complex Cells

Complex cells

- Further processing pooling information from simple cells
- ***Phase invariance:***
  - Phase ϕ of oriented plane wave does not mater
  - light–dark $\sim$ dark–light
  - Looking for edges at a given scale, orientation
- Found in higher-order (2/3) and output (6) layers of V1, and higher-order visual areas

pause

Wait, what do we mean by layers of V1?

The cortical microcircuit

The cortical microcircuit

The mammalian neocortex is fairly homogeneous: most parts of it look similar (with important exceptions). Some aspects of V1 architecture apply to cortex in general

6 layers

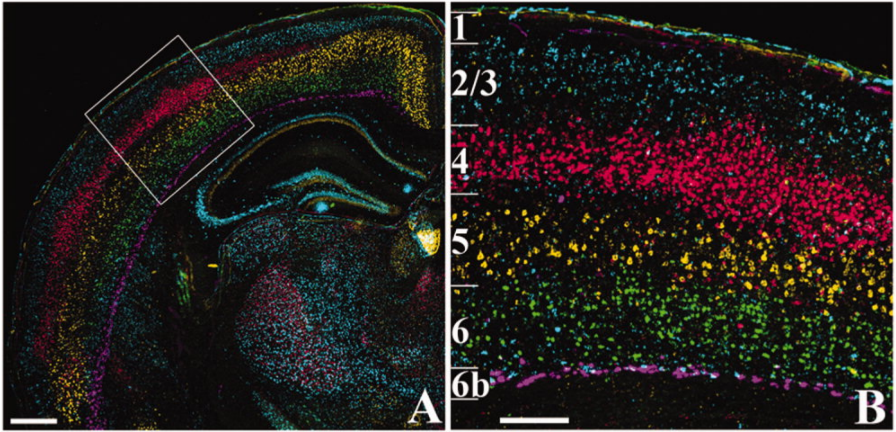
- Numbered from 1–6 starting at the brain surface
- layer 1 is the most superficial
- layer 6 is the deepest

Each layer has a mix of distinct excitatory and inhibitory cell types

- More excitatory neurons than inhibitory neurons
- $\approx 80/20\%$ E/I split
- More types of inhibitory (10–20) than excitatory (4–6)

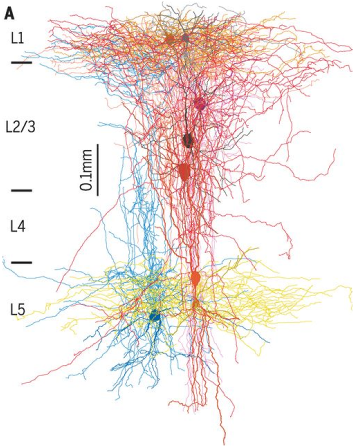
Information flow:

- Thalamus $\rightarrow$ layer 4 $\rightarrow$ layer 2/3 $\rightarrow$ other parts of the brain and layer 5
- Layer 5: outputs (usually motor-like)
- Layer 6: Feedback modulation of cortico-thalamic loops

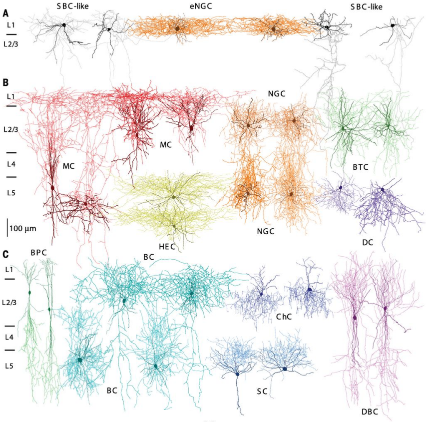


Boyle et al. (2011)

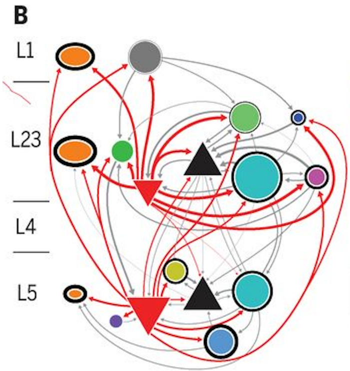
Excitatory (glutamatergic) pyramidal cells



Inhibitory (GABAergic) interneurons



Circuit Diagram



Jiang et al. (2015)

pause

6-layered neocortex (oversimplified)

- Layer 4: _____ from thalamus, e.g. retinal signals arrive in V1 via LGN
- Layer 2/3: Local recurrent processing, sends signals within _____
- Layer 5: Sends motor-like or motor-related outputs
- Layer 6: Closes thalamic feedback loop

V1 tuning curves

- Simple cells
  - Found in ________________ that receive inputs from the LGN (part of the thalamus)
  - Pool together multiple centre-surround inputs to detect _____________
  - Tuning ≈ ______________: Oriented plane-wave tuning × Gaussian receptive field
  - _______ of plane wave matters: Cell may respond to light-then-dark but not dark-then-light

- Complex cells
  - Pool inputs over multiple simple cells
  - ______________: can respond to both light-then-dark and dark-then-light
  - Found in ____

6-layered neocortex (oversimplified)

- Layer 4: Input from thalamus, e.g. retinal signals arrive in V1 via LGN
- Layer 2/3: Local recurrent processing, sends signals within cortex
- Layer 5: Sends motor-like or motor-related outputs
- Layer 6: Closes thalamic feedback loop

V1 tuning curves

- Simple cells
  - Found in V1 layers 4 and 6 that receive inputs from the LGN (part of the thalamus)
  - Pool together multiple centre-surround inputs to detect oriented edges
  - Tuning $\approx$ Gabor function: Oriented plane-wave tuning $\times$ Gaussian receptive field
  - "Phase" of plane wave matters: Cell may respond to light-then-dark but not dark-then-light

- Complex cells
  - Pool inputs over multiple simple cells
  - Phase invariant: can respond to both light-then-dark and dark-then-light
  - Found in V1 layers 2/3 and 6, and higher-order visual areas

end

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